

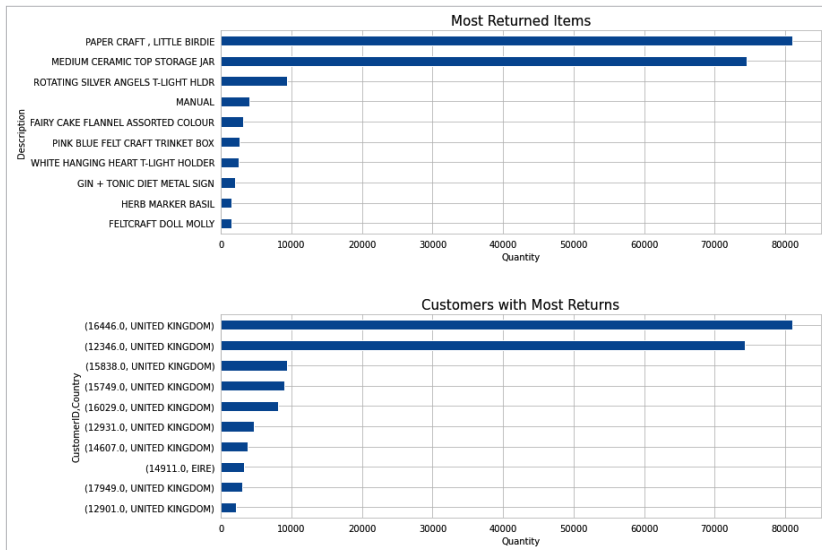


Figure 4: Most returned items and the customers with the corresponding country

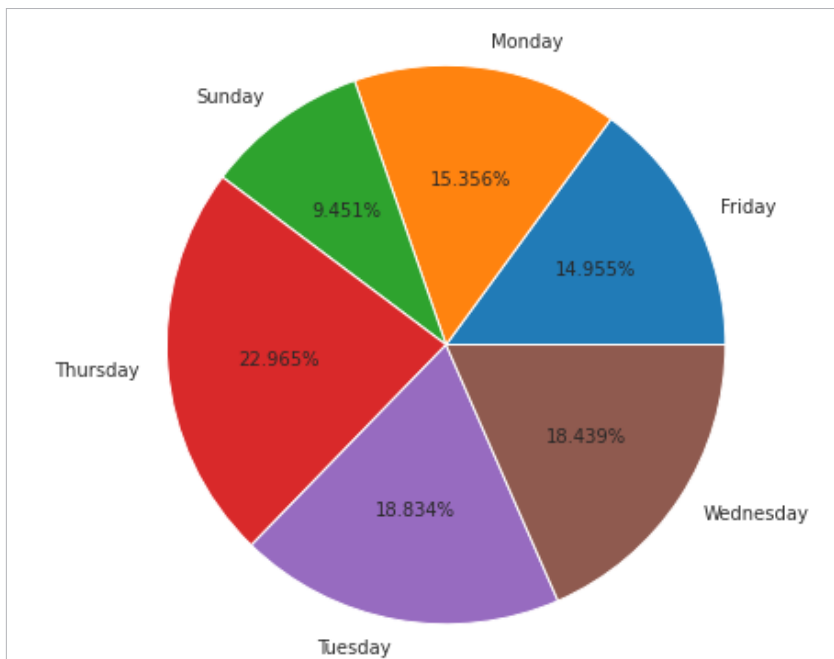


Figure 5: Sales value by day of the week

```
fontsize=15)
ax1.set(xlabel='Quantity')
ax2.set(xlabel='Quantity')
plt.subplots_adjust(hspace=0.4)
plt.show()
```

Figure 4 shows the graph plotted for the above criteria.

Since we have got the day of the week in which the things were sold, we can use it to see the sales value by each

day of the week. We can create the pie chart as shown in Figure 5.

```
#creating the pie chart
Cleaned_Data.groupby('Day of week')
['FinalPrice'].sum().plot(kind = 'pie',
autopct = '%.3f%%', figsize=(7,7)).
set(ylabel='')
plt.title('Percentages of Sales Value
by Day of Week', fontsize = 17)
plt.show()
```

Model building

Let's first save the data set to be used in an Excel file, so that we can use it according to our needs.

```
#Creating an Excel File for the Cleaned
Data
Cleaned_Data.to_excel('Online_Retail_
data.xlsx')
```

```
#Loading the excel file cleaned data
into a dataframe
final_data=pd.read_excel('Online_Retail_
data.xlsx',index_col=0)
final_data.head()
```

```
# convert the StockCode to string
datatype
final_data['StockCode']= final_
data['StockCode'].astype(str)
```

```
# List of unique customers
customers = final_data["CustomerID"].
unique().tolist()
len(customers)
```

The output will be:

4371

Our data set contains 4,371 customers. We will extract the purchasing history of each of these clients. In other words, we can have 4,371 different purchase sequences.

To construct word2vec embeddings, we'll leverage data from 90 per cent of our clients. The remaining data set will be utilised for validation.

Let's split the data set.

Train - test data preparation

```
# shuffle customer ID's
random.shuffle(customers)

# extract 90% of customer ID's
customers_train = [customers[i] for i
in range(round(0.9*len(customers)))]

# split data into train and validation
set
```